

TROUBLE SHOOTING GUIDE



Warning

Always disconnect power source (lockout/tagout) to machine before proceeding with any maintenance, adjustments or repair. Failure to disconnect power may cause serious personal injury and/or damage to machine.



Warning

This machine is equipped with a very sharp knife and powerful laser. Use caution when working on this machine. Failure to keep hands, arms, and loose objects away from knife area may cause serious personal injury.

Service and maintenance to this machine should be performed by qualified personnel. If you do not have qualified personnel, contact your Eastman Sales Representative or Eastman Factory direct.

Problem: UIT Does Not Power Up

Description of Problem:

The UIT does not power. Screen is blank and not lit up.

Troubleshooting:

1. Touch the screen to see if it turns on. If the machine sits idle for a long period of time the touch screen goes into a sleep mode to protect the screen.
2. Make sure your cutting software is open on the cutting system computer.
4. Find the black ethernet power supply (PoE) located next to the computer. Verify the green LED is illuminated.
5. If there is no green LED lit, check the AC outlets located on the back of the diagnostic cabinet. If there is no AC power, check the AC power cord and fuses F3 and F4 in the diagnostic cabinet.

Problem: Machine Stop during Cut Due to Unintentional pause

Description of Problem:

The machine stops in middle of cut and displays message "Machine Paused, Press Zero, Next or Abort" on Touch Screen. When pressing NEXT on the keypad, the machine will continue to cut where it left off. This is typically caused by an intermittent pause circuit, usually in the stop discs.

Troubleshooting:

1. Check stop disc activation. The stop discs should not activate by a slight touch or vibration. They should activate only when moved by minimum pressure. Remove gantry side cover. Check the pause plunger. Tighten the plunger to increase activation pressure and loosen to decrease activation pressure.

Problem: The buttons on the Touch Screen are out of alignment

Description of Problem:

The touch screen buttons do not match where the screen needs to be touched to activate the command. Operator needs to push above, below, right or left of the button for the command to take effect.

Troubleshooting:

Touch screen calibration is required. Proceed as follows.

- 1) Power down the cutting system.
- 2) Place finger on the upper left corner of the touch screen as Shown. While holding finger on the screen, turn ON the main power switch located on the back of the diagnostic control cabinet.



- 3) Remove finger after the "Power On Setup" screen appears.
- 4) Using the left and right arrows to navigate, go to the Calibration Screen (screen page 2). Press Touch screen to begin calibrating.



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- 5) Using a paper clip or fine tip object, press the center of the cross as shown. The cross will reappear in the lower right corner. Press the center of the cross.



- 6) Navigate to screen page 3. Select save and exit to complete calibration.

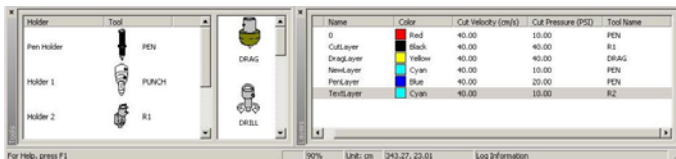
The touch screen will restart in normal mode. Check for proper screen operation and begin your normal start-up procedure.

Problem: The Cutting Tool or Pen does not move down

Description of Problem:
The tool or Pen does not move down when cutting a file or they are delayed coming down at the beginning of a cut. This can be caused by an electrical short, tool mapping in software or a problem with the power supply.

Troubleshooting:

1. Make sure the gantry Tools ON switch is turned on. This can be verified by making sure the laser pointer is on.
2. Verify the mapping of your tools and layers are mapped properly under each tool.



- a) Look at the tool bar at the bottom of the Cut window to quickly verify tool and layer mapping. If there is no Tool Bar as shown, then click on "View" then click on "Layers" and "Tools" in the main E-Suite menu.
- b) If the layer is not mapped to a tool or the tool is not on the tool holder then just click and drag the tool to the spindle or the layer to the tool before sending the file to the cutter.

3. With E-Suite closed, check tool holder to see if it moves freely and is not damaged or bent.

4. Make sure you have pressure at the tools. Try pulling tool the holder down by hand to see if it has pressure.

5. Check tool connections.

- a) Hit the Cut Down button on the UIT to verify the corresponding green LED light is on. Located on the slice output cards (non-operator gantry side plate).

- b) Try firing the tool manually by pressing red button under solenoid block. Each solenoid can be manually fired by pressing the individual button.

- c) Check cable connection at air manifold for proper connection.

6. The Pen or Tool delays coming down and misses the beginning of marker or cut.

- a) Check the 24 VDC power supply located in the diagnostic cabinet. Make sure the green led is on.

- b) Check the 24 VDC power supply fuse.

- c) Check flow control valves on the pen air cylinder.

Problem: Laser Fume Extractor has reduced vacuum or no vacuum.

Description of Problem:

An odor is present when cutting with the laser tool or Laser cutting path is wider than normal

Troubleshooting:

1. Check the laser fume extractor for power and is turned ON.
2. Increase vacuum pressure using the fume extractor control panel.
3. Replace vacuum filters.
4. Clean vacuum hose and funnel.

Problem: Laser cutting path is too wide**Description of Problem:**

The laser cutting path increases to an undesirable width.

Troubleshooting:

1. Check for damage to brass laser nozzle. Replace nozzle if any damage is present.
2. Check for brass laser nozzle alignment. If the tube or nozzle is bent, the laser beam will reflect off the inside wall of the nozzle causing a wide path. To verify, perform a test pulse on a piece of card board. If the dot has a ring or partial ring around it, the nozzle is bent or the mirrors may need to be adjusted.
3. Check laser power and cutting speed. Settings may vary when changing materials. Testing will yield best results.

RF/EMI Interference

Some factory environments may have equipment that generates Radio Frequency (RF) or Electro-Magnetic Interference (EMI). These signals in close proximity to the Eastman cutting system can generate electrical noise and cause problems for the machine and computer (Eastman does offer a shielded mouse). It is recommended that any RF Welders or other equipment generating RF or EMI noise be a minimum of 75 feet (23 meter) from the Eastman cutting system.

Problems of this nature are typically difficult to resolve and must be approached in a systematic manner. To reduce the effects of electrical noise generated by RF, Eastman Machine Company has the following recommendations:

Step 1) Install an earth ground as close as possible to the transformer feeding AC power to the cutting system.

Step 2) Enclose all AC power cables in rigid conduit and secure to the earth ground. Route the four power cables for the computer controls around the end of the conveyor instead of under the unit.

Step 3) Ground conduits to the table frames.

Step 4) Implementation of the foregoing should reduce the level of interference to a workable level. In the event that RF is still determined an issue, installation of RF filters and/or power line filters may be required.

Problem: Loss of laser cutting power**Description of Problem:**

The laser cutting power decreases causing non-cut edges.

Troubleshooting:

1. Damaged or dirty lens. Check for damage or dirt on the focusing lens. Clean or replace lens as required.
2. Damaged or dirty Mirror. Check for damage, burn mark or dirt on the beam deflecting mirrors. Clean or replace mirrors as required.
3. Damage or debris in laser nozzle. Check for damage or debris in laser nozzle. Clean or replace laser nozzle as required.

ELECTRICAL & PNEUMATIC DIAGRAMS

The following electrical and pneumatic drawings are for reference only. Eastman maintains the right to change electrical and pneumatic specifications without notice. Any modifications to machine wiring without written permission from Eastman Machine Company shall void all warranties.